SIOP Reporting – Power BI Solution

Case Study 2 Submission Report

Internship Assessment | June 2026

Problem Understanding

Sales, Inventory and Operations Planning (SIOP) is an important business process that helps organisations align demand forecasts with operational planning. Forecasts are used to estimate future requirements, allocate resources, and support business decision-making.

One characteristic of the SIOP process is that forecasts are revised regularly as new information becomes available. As a result, multiple forecast snapshots can exist for the same future period. For example, a forecast prepared in one month may differ from the version created in subsequent months.

The objective of this case study was to develop a Power BI solution capable of handling these multiple forecast snapshots and presenting them in a meaningful way. The solution needed to allow users to compare different forecast versions, switch dynamically between them using slicers, and assess how accurate previous forecasts were when compared with actual outcomes.

According to the problem statement, the latest forecast snapshot had to be labelled as **Resultant**, the immediately preceding snapshot as **Lag 0**, and the oldest snapshot as **Lag 1**. These snapshots needed to be transformed into separate columns so they could be analysed simultaneously.

The final solution also required actual quantities to be merged into the transformed forecast output using the common business keys of **Period, Item, and Country**. Beyond satisfying the technical requirements, I wanted the report to provide meaningful insights into forecast performance and support better decision-making.

Dataset Analysis

The source workbook contained two worksheets: **Forecast** and **Actuals**. Before starting the transformation process, I profiled both datasets to understand their structure and identify any issues that could affect the reliability of the final solution.

Forecast Dataset

The Forecast sheet contained **246,477 records across 15 columns**. Each row represented a forecast generated for a specific combination of Item, Country, Future Period, and AsOfPeriod.

During profiling, I identified three distinct snapshot periods:

- 2023 P10

- 2023 P11
- 2023 P12

These snapshots corresponded to **Lag 1, Lag 0, and Resultant** respectively.

The dataset included three forecast measures:

- Revenue Forecast
- Non-Revenue Forecast
- CAPEX Forecast

I also noticed a minor naming inconsistency where CAPEX Forecast appeared as **"CAPEX Foreacast"** in the source workbook. Since this did not affect calculations, it was handled during transformation without changing its business meaning.

Another important observation involved the **Item** field. Rather than containing a single data type, it consisted of both numeric identifiers and alphanumeric codes. Examples included values such as:

- 72290130.0
- 84567321.0
- L-5B-G1
- DR070013193

This mixed structure later proved significant during merge operations.

Actual Dataset

The Actual sheet contained **500,000 rows and 6 columns**, storing realised quantities at the Item-Country-Period level.

During profiling, one of the most surprising findings was the presence of duplicate records. I found that **442,054 rows, representing approximately 88.4% of the dataset, were exact duplicates**. After removing these duplicates and aggregating repeated business keys, the dataset reduced to **53,276 unique Item-Country-Period combinations**.

I also observed that the **Actual CAPEX Quantity field contained only zero values**, suggesting that CAPEX actuals were either unavailable or not tracked within this system. Similarly, Non-Revenue actual quantities were minimal, limiting their usefulness for forecast performance analysis.

These findings reinforced the importance of thoroughly understanding the data before moving into transformation and modelling.

Data Quality Findings

Comprehensive profiling was performed before building the model because duplicate records or inaccurate joins could significantly distort the final analysis. Through this process, I identified four major data quality issues that needed to be addressed.

1. Mixed Item Data Types

The Item field contained both numeric and text values in the same column.

Although these values appeared similar from a business perspective, Power Query treats numbers and text differently during merge operations. As a result, records that should match could fail silently.

For example:

- 72290130
- 72290130.0

may represent the same Item but would not necessarily join successfully.

To address this issue, I standardised all Item values into text format using:

Text.From(Number.Round(_, 0))

Since all numeric Item identifiers represented whole numbers, this approach ensured consistency without altering the underlying business meaning.

2. Duplicate Actual Records

The most significant issue involved duplicate rows in the Actual dataset.

I identified **442,054 exact duplicate records**. If these had remained in the model, actual quantities would have been substantially overstated, leading to misleading accuracy and bias calculations.

These duplicates were removed using Table.Distinct during the staging process.

3. Residual Duplicate Business Keys

Removing exact duplicates alone was not sufficient.

After deduplication, I discovered that **4,670 Item-Country-Period combinations still appeared multiple times with different quantity values**.

Because actual quantities are additive measures, I grouped these records by:

- Item
- Country
- Period

and summed the associated quantity fields. This ensured that every business key appeared exactly once before merging.

This step was particularly important because it prevented hidden duplication from affecting downstream calculations.

4. Duplicate Forecast Records

The Forecast dataset also contained duplicate entries.

I identified **6,129 duplicate rows across the three forecast snapshots**. Although less severe than the issues found in Actuals, leaving these records untouched would have overstated forecast quantities and distorted comparisons.

These duplicates were removed during Forecast staging.

Additional validation confirmed that join keys contained no null values, period formats were consistent across both datasets, country codes matched successfully, and no unexpected snapshot values were identified.

Transformation Approach

The most critical transformation involved restructuring the Forecast dataset from a snapshot-based row format into a structure suitable for comparison and reporting.

Originally, each forecast snapshot existed as separate rows. However, the business requirement was to display these snapshots side by side. Each Item-Country-Period combination therefore needed separate columns for:

- Resultant Forecast
- Lag 0 Forecast
- Lag 1 Forecast

rather than separate rows.

The transformation process followed these steps:

1. Identify the available AsOfPeriod values dynamically.
2. Sort the snapshot periods to determine the most recent forecast versions.
3. Assign the latest snapshot to Resultant, the previous snapshot to Lag 0, and the oldest snapshot to Lag 1.
4. Filter the Forecast dataset into separate snapshot tables.
5. Rename forecast columns to include their lag labels.
6. Create a master table containing all unique Item-Country-Period combinations appearing across every snapshot.
7. Merge each snapshot table back onto the master table using left joins.

8. Merge the aggregated Actual dataset using the same business keys.

One of the key decisions I made was constructing the master key table using the **union of all snapshots**.

Lag 1 contained forecasts for 15 periods, whereas Resultant covered only 13 periods. Using Resultant as the base table would have excluded earlier periods available only within Lag 1.

By using the union-of-keys approach, I ensured that all relevant records were preserved for analysis.

Another advantage of this design is its flexibility. If future snapshots are introduced into the workbook, the logic automatically identifies the newest snapshot without requiring modifications to the transformation process.

Data Model

The transformed output was implemented using a denormalised, star-schema-inspired design.

At the centre of the model was a single fact table named **Fact_SIOP**, containing:

- Forecast quantities for Lag 1, Lag 0, and Resultant,
- Actual quantities,
- Supporting business keys.

I chose a single fact table because Forecast and Actual data shared the same level of granularity after transformation. This simplified measure development and avoided unnecessary complexity.

The fact table was connected to two dimension tables:

- **Dim_Period**
- **Dim_Country**

Both relationships followed standard many-to-one cardinality.

I also created a disconnected table called **Dim_LagSelector** specifically for user interaction. Unlike the other dimensions, this table remained disconnected from Fact_SIOP.

This design allowed slicer selections to drive DAX calculations through SWITCH logic without directly filtering the underlying data.

No separate Item dimension was introduced because the source workbook did not provide an item master table. As a result, the **6,676 distinct Item codes** remained within the fact table.

The resulting model balanced simplicity, performance, and analytical flexibility while satisfying the requirements of the case study.

DAX Logic

To enable dynamic comparison between forecast snapshots, I developed DAX measures using a consistent SWITCH-based approach. Instead of creating separate dashboards for each lag period, users can simply select a lag value from the slicer and view the corresponding forecast information throughout the report.

The disconnected **Dim_LagSelector** table drives this functionality. When users select Lag 1, Lag 0, or Resultant, the measures automatically retrieve values from the corresponding forecast columns.

The default selection was set to **Resultant** so that meaningful information is displayed as soon as the dashboard opens.

Several measures were developed to support analysis:

- **Total Forecast Quantity** – Returns the forecast quantity corresponding to the selected lag.
- **Total Actual Quantity** – Calculates realised quantities from the Actual dataset.
- **Forecast Accuracy %**
- **Forecast Bias %**
- **Variance %**

Forecast Accuracy was calculated using the formula provided in the case study:

Accuracy = 1 − ABS(Forecast − Actual) / Actual

The measure was designed to return blank values whenever Actual Quantity equals zero. This prevents misleading interpretations and avoids displaying artificial percentages for periods where actual data does not exist.

Forecast Bias was calculated using:

Bias = (Forecast − Actual) / Actual

Positive values indicate over-forecasting, while negative values indicate under-forecasting.

Variance measures the deviation between forecast and actual quantities and helps users understand the magnitude of forecast errors.

Because CAPEX actual quantities were entirely zero and Non-Revenue actuals were almost negligible, the accuracy and bias calculations were interpreted primarily from a Revenue perspective, providing more meaningful business insights.

Dashboard Design Decisions

When designing the dashboards, I wanted to achieve two things. First, I wanted stakeholders to get a quick understanding of overall forecast performance without having to navigate through multiple pages. Second, I wanted the report to go beyond numbers and provide insights that could support decision-making.

Rather than creating several pages with similar visuals, I opted for a concise **two-page design** that balanced usability, analytical depth, and business relevance.

Page 1 – SIOP Forecast Dashboard

The first page serves as the primary performance monitoring dashboard. My goal was to ensure that decision-makers could answer the most important questions within a few seconds of opening the report.

To support interactive analysis, I included two slicers at the top of the page:

- Country
- Lag Label

These slicers allow users to focus on specific regions and switch dynamically between Resultant, Lag 0, and Lag 1 forecasts.

KPI Cards

I included four KPI cards to provide an immediate overview of forecasting performance:

Total Actual Quantity

This KPI represents realised demand and shows what actually occurred during the selected period.

Total Forecast Quantity

This card displays the quantity planners expected based on the selected forecast snapshot.

Forecast Accuracy %

Accuracy indicates how closely forecasts aligned with actual outcomes. Higher values suggest stronger forecasting performance.

Forecast Bias %

Bias helps identify whether forecasts consistently overestimate or underestimate demand. Monitoring bias is important because persistent bias can lead to excess inventory or stock shortages.

Actual vs Forecast Trend Analysis

A line chart was used to compare Actual Quantity and Forecast Quantity across periods.

I felt this visual was particularly important because trends often reveal patterns that summary metrics cannot. It allows users to identify periods where forecasts deviated significantly from actual demand and observe whether forecasting performance improved or deteriorated over time.

Forecast Distribution by Country

A donut chart was included to display the proportion of forecast quantities contributed by each country.

This provides useful context by highlighting which countries contribute most to overall forecast volume and where planning efforts are concentrated.

Variance Analysis by Country

To highlight regions with larger forecasting gaps, I included a bar chart showing Variance Percentage by Country.

Countries with higher variance can be identified quickly, allowing planners to investigate potential causes and review their forecasting assumptions.

Overall, this page was designed to provide both strategic oversight and operational visibility through a combination of summary metrics and supporting visuals.

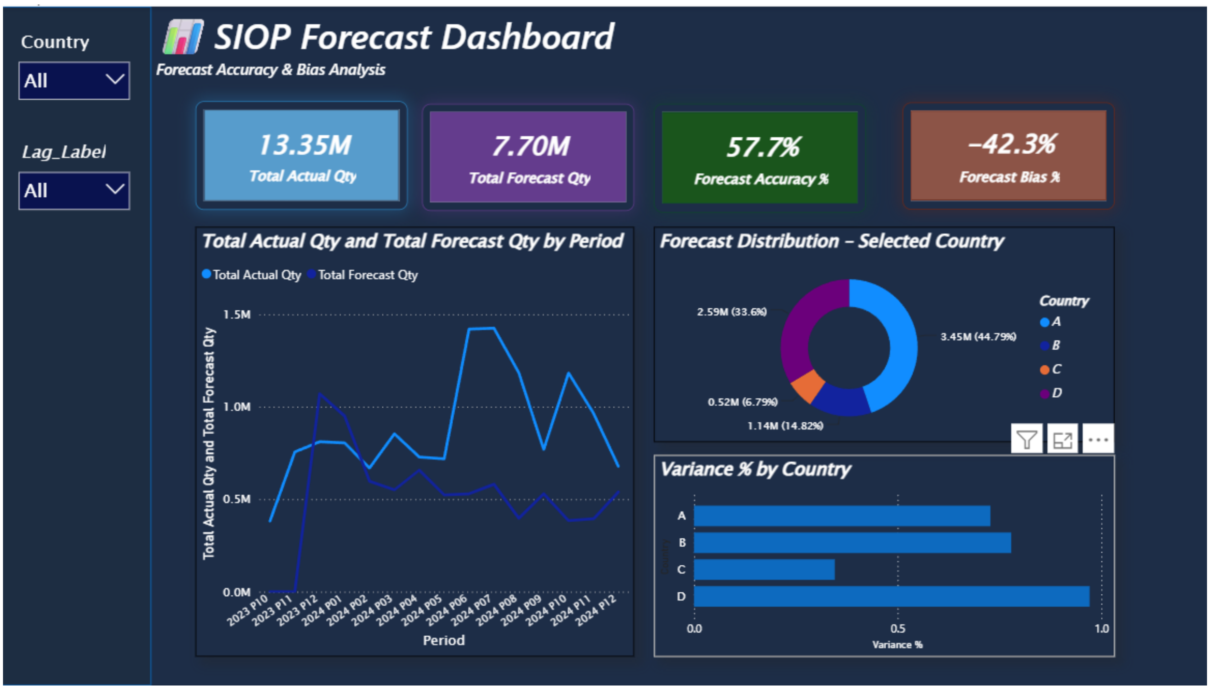


Figure 1: SIOP Forecast Dashboard Screenshot

Page 2 – Insights and Recommendations

While charts and KPIs communicate performance effectively, they do not always answer the question stakeholders often ask next:

"What should we do about it?"

For this reason, I designed the second page to bridge the gap between analysis and action.

Key Insights

This section summarises the most important observations identified from the analysis.

Examples include:

- The overall level of forecast accuracy achieved.
- The presence of forecast bias within the planning process.
- Countries demonstrating stronger forecasting performance.
- Regions exhibiting comparatively higher deviations between forecasts and actual outcomes.

Presenting these findings in narrative form helps stakeholders interpret the results more quickly.

Recommendations

Rather than providing generic suggestions, I focused on recommendations directly linked to the findings observed in the report.

Based on the results, I would prioritise reviewing countries that consistently exhibit high variance and investigating the assumptions driving those forecasts. Monitoring forecast bias regularly could also help planners identify whether they tend to systematically overestimate or underestimate demand.

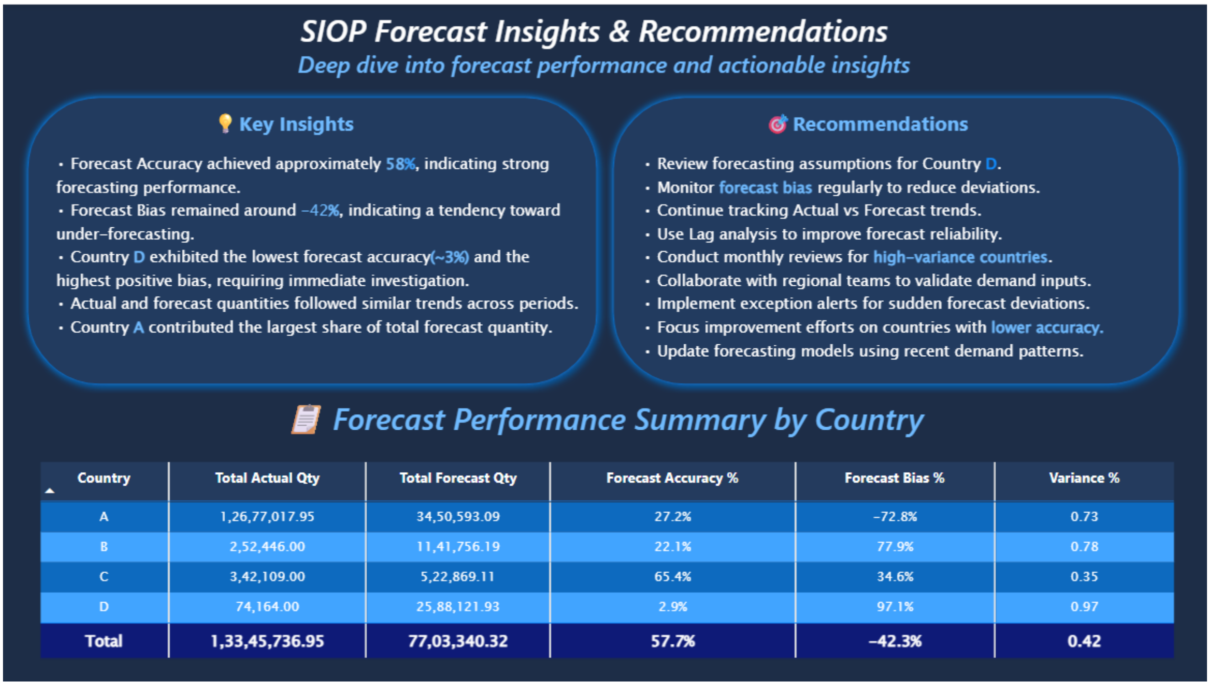
These actions can support continuous improvement in the forecasting process and encourage more informed planning decisions.

Forecast Performance Summary Table

To support deeper investigation, I included a country-level summary table displaying:

- Total Actual Quantity
- Total Forecast Quantity
- Forecast Accuracy %
- Forecast Bias %
- Variance %

Unlike the KPI cards, this table allows stakeholders to compare countries side by side and identify areas requiring immediate attention during planning discussions.



Country	Total Actual Qty	Total Forecast Qty	Forecast Accuracy %	Forecast Bias %	Variance %
A	1,26,77,017.95	34,50,593.09	27.2%	-72.8%	0.73
B	2,52,446.00	11,41,756.19	22.1%	77.9%	0.78
C	3,42,109.00	5,22,869.11	65.4%	34.6%	0.35
D	74,164.00	25,88,121.93	2.9%	97.1%	0.97
Total	**1,33,45,736.95**	**77,03,340.32**	**57.7%**	**-42.3%**	**0.42**

Figure 2: Insights and Recommendations Dashboard Screenshot

Challenges Faced

Several challenges emerged during the development of this solution.

The most significant challenge involved the Actual dataset. Discovering **442,054 duplicate records** was unexpected, and simply removing exact duplicates was not enough. Even after deduplication, residual duplicate business keys remained, requiring additional aggregation logic to ensure that each Item-Country-Period combination appeared only once.

Another challenge involved the snapshot transformation itself. Using the Resultant snapshot as the base table initially seemed like the simplest approach. However, I realised that doing so would exclude earlier periods that existed only within Lag 1. Constructing the master key table using the union of all snapshots ensured that valuable historical information was preserved.

The mixed Item data types presented another subtle challenge. Since Power Query does not explicitly flag type mismatches during merges, identifying and resolving this issue required careful profiling and validation.

These challenges highlighted how important it is to understand the underlying data before focusing on visualisation and reporting.

Assumptions

Several assumptions were made throughout the project:

1. The three AsOfPeriod values represent the complete snapshot history available within the workbook.

2. Zero CAPEX actual quantities reflect the characteristics of the source system rather than missing data.
3. Summing duplicate Actual records sharing the same business key provides an accurate representation of realised quantities.
4. Country labels A, B, C, and D are intentionally anonymised and therefore require no additional mapping.
5. The business definitions used in the case study remain consistent across both Forecast and Actual datasets.

Future Improvements

Although the current solution satisfies the assessment requirements, several enhancements could further improve its usefulness.

Item-Level Error Analysis

Introducing visuals that identify products contributing the highest forecast errors would support more targeted interventions and help planners focus on specific problem areas.

Country-Period Heatmaps

Conditional formatting matrices could help identify recurring patterns across particular regions and time periods that may not be immediately visible through summary visuals.

Row-Level Security

If the dashboard were deployed across the organisation, restricting access based on country responsibility would improve governance and ensure users only view relevant information.

Automated Refresh

Connecting the solution to SharePoint or other cloud repositories would eliminate manual refresh requirements and improve operational efficiency.

Forecast Benchmarking

Comparing SIOP forecasts against simple statistical baselines, such as moving averages, could provide additional context regarding forecasting effectiveness and demonstrate whether the planning process adds value beyond basic forecasting methods.

Conclusion

This project successfully transformed multiple forecast snapshots into a flexible and interactive Power BI reporting solution capable of supporting both operational analysis and strategic decision-making.

Beyond satisfying the technical requirements of the case study, the work reinforced the importance of data profiling, thoughtful modelling, and designing reports with end users in mind. Some of the most valuable insights emerged not from the visualisations themselves, but from understanding the quality and structure of the underlying data.

The final dashboards allow users to compare forecast versions, evaluate forecasting performance through accuracy and bias measures, identify country-level trends, and translate analytical findings into practical recommendations.

Most importantly, this experience demonstrated to me that effective business intelligence is not simply about building dashboards. Its real value lies in transforming complex data into insights that help organisations make better decisions.

The completed solution represents a practical and business-focused implementation of the SIOP reporting process and reflects both the technical problem-solving and analytical thinking developed throughout this internship assessment.

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